INCH-POUND

MIL-T-27/285A
13 March 1991
SUPERSEDING
MIL-T-27/285
17 JANUARY 1980

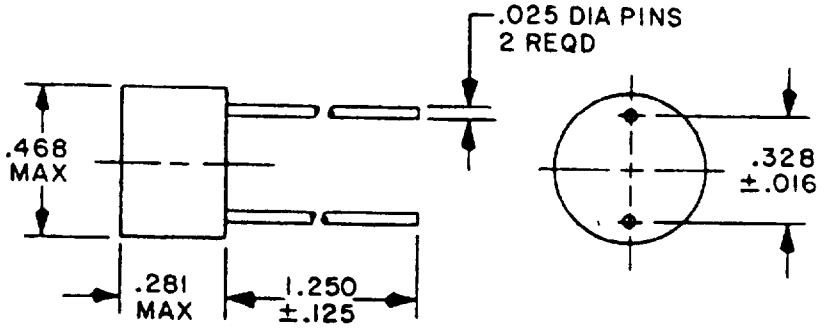
MILITARY SPECIFICATION SHEET

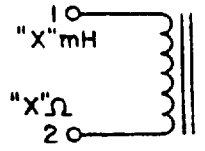
TRANSFORMERS AND INDUCTORS (AUDIO, POWER, AND HIGH-POWER PULSE), INDUCTORS, AUDIO FREQUENCY, HIGH Q, TF5R20ZZ

Ⓐ Inactive for new design after the date of this specification.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-T-27.





WV – 175 V
ALTITUDE – 75,000 FT MAX
(SEE TABLE I FOR VALUE OF "X")

CIRCUIT DIAGRAM AND MARKING

FIGURE 1. Dimensions and configurations.

Ⓐ denotes changes

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Inches	mm
.016	0.41
.025	0.64
.125	3.18
.281	7.14
.328	8.33
.468	11.89
1.250	31.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the side and top of the case.
4 Electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the windings.)

Electrical ratings: See table I.

TABLE I. Electrical ratings.

Dash number No.	Inductance 1/ mH (1-2) ±2 percent	DC 2/ current mA (1-2) max.	DC resistance ohms (max) (1-2)	Quality factor (min)	Temperature stability (-55°C to +105°C)	Working voltage (peak) volts	Voltage (1-2)
01	100	7.0	70	36 at 9 kHz	±1 percent	175	0.1 volt at 1 kHz
02	150	6.0	110	36 at 9 kHz	±1 percent	175	0.1 volt at 1 kHz
03	250	4.5	180	36 at 9 kHz	±1 percent	175	0.1 volt at 1 kHz
04	350	3.7	265	38 at 9 kHz	±1 percent	175	0.1 volt at 1 kHz
05	500	3.0	360	35 at 9 kHz	±1 percent	175	0.1 volt at 1 kHz
06	750	2.5	540	30 at 9 kHz	±1 percent	175	0.1 volt at 1 kHz

1/ The inductance is measured with 0 A dc applied to (1-2) and at the specified voltage across (1-2).
2/ The amount of DC current shall reduce the inductance a maximum of 7 percent.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: continuous.

Case: Diallyl phthalate.

Terminals: 0.025 solderable pins.

Weight: 0.08 ounce maximum.

Altitude: 75,000 feet maximum.

Operating temperature range: -55°C to +105°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 2 pounds

Dielectric withstanding voltage:

At sea level: 500 volts rms.

At reduced barometric pressure: 300 volts rms.

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Vibration (high frequency): MIL-STD-202, method 204.

Ⓐ Quality assurance provisions:

Qualification inspection: Not applicable.

Quality conformance inspection: Groups A and B tests of MIL-T-27 shall be applicable.

Marking location: See figure 1.

Part number: M27/285-(dash number from table I.)

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS, SH
Air Force - 17, 99
DLA - ES

User activities:
Army - AR
Navy - AS, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5950-0753-47)

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